CORPORATE NETWORK PROJECT
TECHSTORE PRO

VITOR ANDRADE

2026

Corporate Network Project – TechStore Pro

1. Introduction

This project aims to design and simulate a corporate network using Cisco Packet Tracer. Fundamental computer networking concepts will be applied, including network segmentation, IP addressing, DHCP, NAT, ACL, and security. The proposal is to simulate a real-world company environment, ensuring organization, security, and efficient communication between departments.

2. Objective

The objective of this project is to create a complete corporate network infrastructure, enabling:

- Controlled communication between departments;
- Automatic IP address allocation (DHCP);
- Implementation of security using ACLs;
- Internet access through NAT;
- Publishing internal services via DNAT.

3. Network Scenario

The company TechStore Pro has four main departments:

- Administrative;
- Financial;
- IT;
- Sales.

Each department has its own workstations and must operate on separate networks for security and organizational purposes. The infrastructure will consist of:

- 1 main router responsible for routing and internet access;
- 2 switches for device connectivity;
- 1 internal server for network services;
- Computers distributed across the departments.

The network must allow controlled internal communication as well as external access to the internet.

4. IP Addressing Plan

For network organization, each department has been separated into a specific subnet:

VLAN	SETOR	REDE	GATEWAY
10	Administrative	192.168.10.0/24	192.168.10.1
20	Financial	192.168.20.0/24	192.168.20.1
30	IT	192.168.30.0/24	192.168.30.1
40	Sales	192.168.40.0/24	192.168.40.1

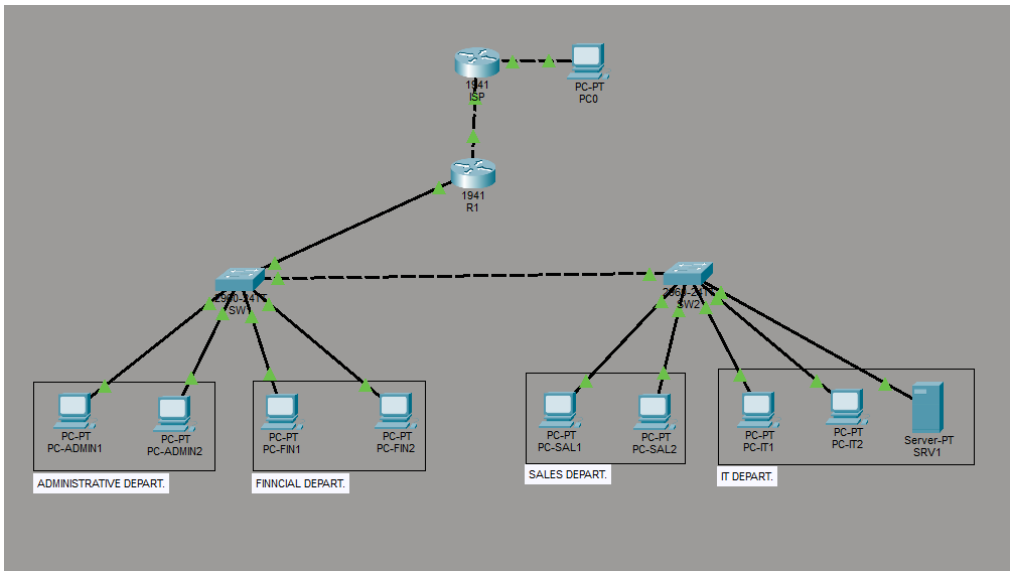
This segmentation improves the network's security and organization.

5. Network Policies

The following rules have been defined for the network:

- Each department will be isolated using VLANs;
- The Sales department will not have access to the Financial department;
- The server will be located within the IT department network;

6. Network Topology



The network topology was built using Cisco Packet Tracer, following a centralized model. The main router was connected to the access switches, which in turn connect the end devices. This structure enables scalability, organization, and logical network segmentation through VLANs.

The devices were distributed across two switches to ensure better organization and network load balancing. Switch 1 connects the Administrative and Financial departments, while Switch 2 connects the Sales and IT departments. The server was connected to Switch 2, as it is associated with the IT department, which is responsible for managing network services.

7. Initial Router Configuration

```
Physical   Config   CLI   Attributes

Router(config)#hostname R1
R1(config)#enable secret ROUTER123
R1(config)#line console 0 password console123
                                   ^
% Invalid input detected at '^' marker.

R1(config)#line console 0
R1(config-line)#password console123
R1(config-line)#login
R1(config-line)#exit
R1(config)#line vty 0 4
R1(config-line)#password vty123
R1(config-line)#login
R1(config-line)#exit
R1(config)#service password-encryption
R1(config)#interface g0/0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#enable
R1#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#do wr
Building configuration...
[OK]
R1(config)#write memory
           ^
% Invalid input detected at '^' marker.

R1(config)#do write memory
Building configuration...
[OK]
R1(config)#

Copy   Paste
```

The basic configuration of the router was performed, including:

- Setting the hostname;
- Configuring access passwords (enable, console, and VTY);
- Enabling password encryption;
- Configuring remote access;
- Activating the network interface.

These steps are essential to ensure security and proper operation of the device in a corporate environment.

8. VLAN Configuration

```
Switch(config)#do show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   ADMIN                            active    Fa0/1, Fa0/2
20   FINANCEIRO                       active    Fa0/3, Fa0/4
30   VENDAS                           active
40   TI                               active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch(config)#
```

VLANs were created to segment the network by department, improving security and organization. Each switch port was assigned to a specific VLAN, representing the company's departments. Trunk mode was also configured between the switches, allowing communication between VLANs through the router.

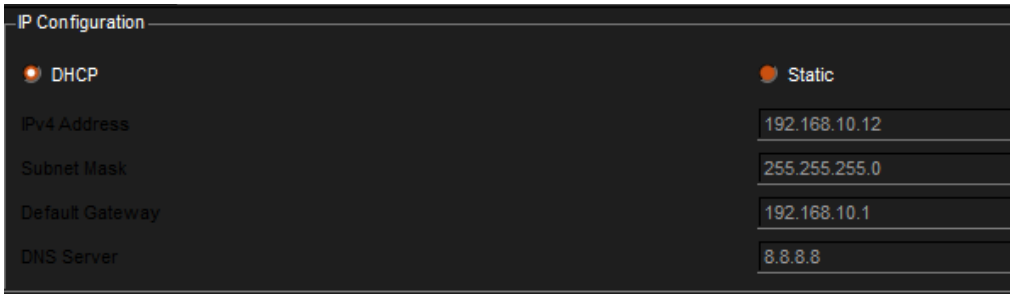
9. Inter-VLAN Routing

```
R1(config-subif)#do show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/0.30  192.168.30.1    YES manual up                    up
GigabitEthernet0/0.40  192.168.40.1    YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1(config-subif)#
```

Inter-VLAN routing was implemented using the Router-on-a-Stick technique. Subinterfaces were created on the router, each associated with a specific VLAN through 802.1Q encapsulation. Each subinterface was assigned an IP address corresponding to the network gateway, enabling communication between the different departments of the company.

During the configuration, an IP addressing error was identified in the gateway of VLAN 10, where an incorrect address was initially used. The issue was resolved by adjusting the address to the correct network range (192.168.10.1), ensuring proper communication between devices. This type of validation is essential in network environments to prevent connectivity issues.

10. DHCP Configuration



The DHCP service was implemented to automate IP address assignments in the network, reducing manual errors and improving scalability. DHCP pools were created for each VLAN, containing the following configurations:

- Network definition (network);
- Default gateway (default-router);
- DNS server;
- Exclusion of reserved addresses.

Each VLAN received a dedicated pool, ensuring organization and control.

DHCP operates through a communication process between client and server, where a device automatically requests an IP address when joining the network. The router was configured as the DHCP server, responding to requests from devices in different VLANs through the previously configured inter-VLAN routing.

During implementation, issues were encountered related to devices not obtaining IP addresses. The analysis identified misconfigurations in the trunk links between switches and in the interface connected to the router, which prevented communication

with the DHCP server. After correcting these configurations, the service functioned properly.

```
Switch(config-if)#
Switch(config-if)#interface fa0/10
Switch(config-if)#switchport mode trunk
Switch(config-if)#no shutdown
Switch(config-if)#do wr
Building configuration...
[OK]
Switch(config-if)#%SPANTREE-2-RECV_PVID_ERR: Received 802.1Q BPDU on non trunk FastEthernet0/5 VLAN1.

%SPANTREE-2-BLOCK_PVID_LOCAL: Blocking FastEthernet0/5 on VLAN0001. Inconsistent port type.


%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

Switch(config-if)#do show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/5       auto         n-802.1q       trunking      1
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/5       1-1005
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Fa0/5       1,10,20,30,40
Gig0/1      1,10,20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/5       1,10,20,30,40
Gig0/1      1,10,20,30,40
```

11. ACL Implementation

An Access Control List (ACL) was configured on the router with the purpose of controlling traffic between the different network departments. The rules were defined according to security policies, allowing full access for Administrative and IT departments, while restricting access for departments such as Finance and Sales to specific resources. The ACL was applied on the router's inbound interface, ensuring traffic control between VLANs.

```
R1(config)#access-list 100 permit ip 192.168.10.0 0.0.0.255 any
R1(config)#access-list 100 permit ip 192.168.40.0 0.0.0.255 any
R1(config)#access-list 100 permit ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
R1(config)#access-list 100 deny ip 192.168.20.0 0.0.0.255 any
R1(config)#access-list 100 deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
R1(config)#access-list 100 permit ip any any
R1(config)#interface g0/0
R1(config-if)#ip access-group 100 in
R1(config-if)#do show access-list
Extended IP access list 100
    10 permit ip 192.168.10.0 0.0.0.255 any
    20 permit ip 192.168.40.0 0.0.0.255 any
    30 permit ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
    40 deny ip 192.168.20.0 0.0.0.255 any
    50 deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
    60 permit ip any any
```

12. NAT Implementation

The objective of this stage was to allow devices within the internal network, using private IP addresses, to communicate with a simulated external network representing the internet. To achieve this, NAT (Network Address Translation) was implemented, responsible for translating internal IP addresses into a public IP address.

A second router was added to the environment, representing an Internet Service Provider (ISP). The communication flow was structured as follows:

Internal Network → Main Router (R1) → ISP Router → Simulated External Network

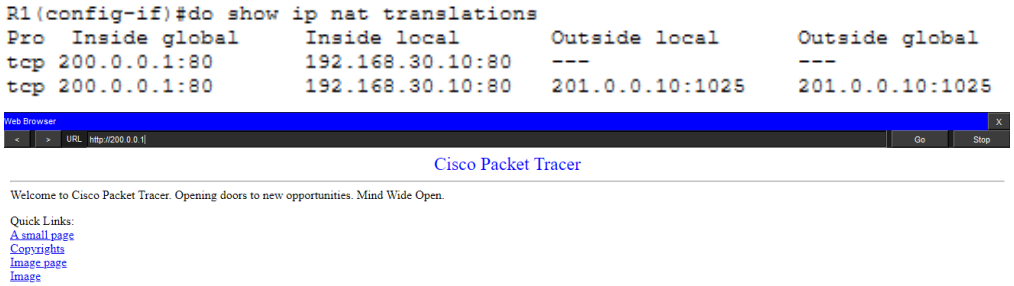
This setup enabled the simulation of a real-world internet access scenario in corporate environments.

Connectivity tests were performed from internal network devices toward the ISP router. The tests confirmed that, after implementing NAT, the devices were able to successfully communicate with the external network. The command "show ip nat translations" was also used to verify dynamic address translation, confirming that NAT was functioning correctly.

13. DNAT Implementation

Destination NAT (DNAT) was implemented to allow external access to a web server located within the internal network. A static NAT rule was configured on the router, mapping the public IP address to the private IP address of the server, specifically for HTTP traffic (port 80). This ensures that requests originating from the external network are correctly redirected to the internal server.

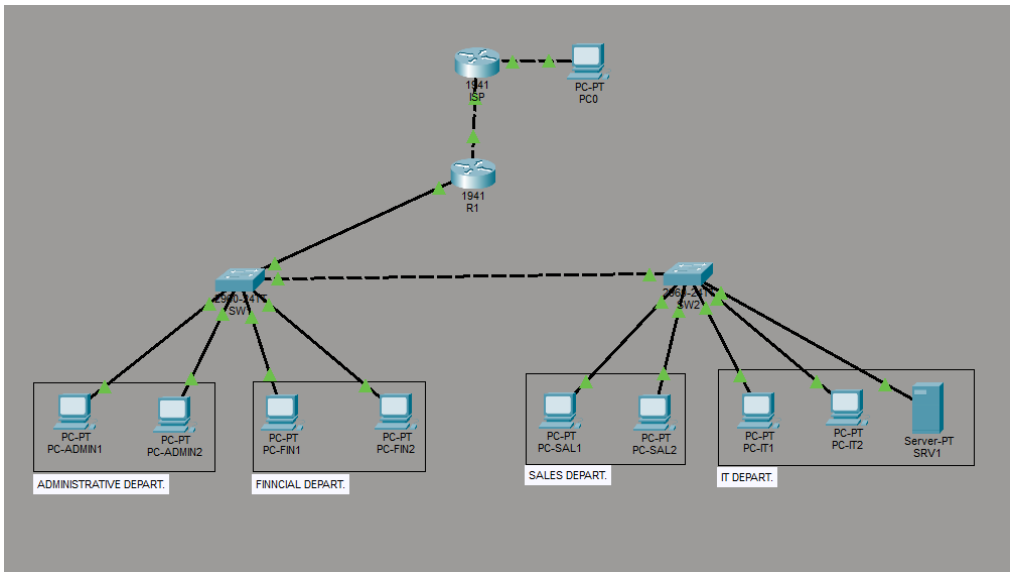
The server was configured with a static IP address and with the HTTP service enabled, simulating the deployment of a web application. Additionally, the internal and external interfaces on the router were properly defined, ensuring the correct operation of the address translation process. Validation was performed from an external device, which successfully accessed the server using the public IP address, confirming that DNAT was functioning correctly.



14. Conclusion

This project successfully implemented a complete network infrastructure using VLAN segmentation, inter-VLAN routing, DHCP, ACLs, NAT, and DNAT. The result allowed internal devices to communicate securely across different network segments, access external networks, and provide internal services to external users.

Throughout the implementation, troubleshooting steps were required to resolve issues related to VLAN configuration, trunking, IP addressing, and routing, reinforcing practical networking skills. This project demonstrates a solid understanding of corporate networking fundamentals and simulates real-world scenarios commonly found in enterprise environments.



15. Network Verification and Troubleshooting Commands

To validate the network configuration and ensure proper operation, several verification commands were used on the router and switches:

- *show ip route*

  Used to verify the routing table and confirm that all VLAN networks are reachable through the router.

```
R1(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 200.0.0.2 to network 0.0.0.0

     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
     192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0.30
     192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.40.0/24 is directly connected, GigabitEthernet0/0.40
L       192.168.40.1/32 is directly connected, GigabitEthernet0/0.40
```

- *show ip nat translations*

  Used to verify NAT operation by displaying active address translations between internal and external networks.

```
R1(config)#do show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
tcp 200.0.0.1:80       192.168.30.10:80   ---                ---
```

- *show access-lists*

  Used to validate ACL rules and ensure that traffic filtering policies are being correctly applied.

```
R1(config)#do show access-lists
Extended IP access list 100
    10 permit ip 192.168.10.0 0.0.0.255 any
    20 permit ip 192.168.40.0 0.0.0.255 any
    30 permit ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
    40 deny ip 192.168.20.0 0.0.0.255 any
    50 deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
    60 permit ip any any
Standard IP access list 1
    10 permit 192.168.0.0 0.0.255.255
```

- *show ip interface brief*

  Used to confirm that all VLAN gateways are correctly configured with the appropriate IP addresses and that their operational status is up/up, ensuring proper inter-VLAN routing.

```
R1(config)#do show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.10.1    YES manual up                    up
GigabitEthernet0/0.20  192.168.20.1    YES manual up                    up
GigabitEthernet0/0.30  192.168.30.1    YES manual up                    up
GigabitEthernet0/0.40  192.168.40.1    YES manual up                    up
GigabitEthernet0/1     200.0.0.1       YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

- *show interfaces trunk*

  Used to confirm that trunk ports are active, identify which VLANs are allowed to pass, and ensure that inter-switch communication and VLAN traffic are properly configured.

```
Switch(config)#do show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/5       auto         n-802.1q       trunking      1
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/5       1-1005
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Fa0/5       1,10,20,30,40
Gig0/1      1,10,20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/5       1,10,20,30,40
Gig0/1      1,10,20,30,40
```

- *show vlan brief*

  Used to confirm that VLANs were correctly created and that ports were properly assigned.

```
show interface brief
                ^
% Invalid input detected at '^' marker.

Switch(config)#do show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/2
10   ADMIN                            active    Fa0/1, Fa0/2
20   FINANCEIRO                       active    Fa0/3, Fa0/4
30   VENDAS                           active
40   TI                               active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch(config)#

Switch#do show vlan brief
         ^
% Invalid input detected at '^' marker.

Switch#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#do show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/8, Fa0/9
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   VLAN0010                         active
20   VLAN0020                         active
30   VLAN0030                         active    Fa0/1, Fa0/2, Fa0/7
40   VLAN0040                         active    Fa0/3, Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

These commands are essential for troubleshooting and validating network behavior in real-world environments.